

CHANIA UWASE

Mechatronics | Automation | Artificial Intelligence

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PROFESSIONAL SUMMARY

Mechatronics engineering graduate with postgraduate training in mechatronics and artificial intelligence, and hands-on experience in electronics prototyping, fabrication, testing, and field deployment. Technical foundation in PLC/HMI programming, CAD, Python, and computer vision. Quick learner with bilingual English/French technical-support experience, seeking junior automation, controls, or mechatronics opportunities.

TECHNICAL SKILLS

Automation and Controls: Programmable Logic Controllers (PLC), Human-Machine Interface (HMI), Studio 5000, RSLogix 500, Siemens TIA Portal, FactoryTalk View, ladder logic, digital and analog I/O, interlocks, alarms, PID control fundamentals

Industrial Networks and Tools: EtherNet/IP, PROFINET, Modbus, OPC UA, Kepware; multimeter and oscilloscope fundamentals

Mechanical Design: SolidWorks, Fusion 360, AutoCAD, assembly and component drawings, sectional views, bills of materials, 3D printing (SLS, PLA, resin)

Embedded Systems and Electronics: STM32, ESP32, C/C++, sensors, board bring-up, SMD soldering and desoldering, hot-air rework, conformal coating

Programming and AI: Python, NumPy, Pandas, SQL, YOLOv8, computer vision, dataset preparation, machine learning experimentation

Technical Support: QuickBooks Online, SaaS troubleshooting, error-code diagnosis, issue reproduction, payment and payroll workflows, data-migration support, Tier 2 escalation, technical documentation

Languages: English, French, Kinyarwanda

PROFESSIONAL EXPERIENCE

Technical Support Representative (Bilingual English/French)

Nov 2022 – Present

Intuit QuickBooks Canada | Remote

- Troubleshoot QuickBooks Online, Payroll, Payments, and QuickBooks Time issues involving account configuration, banking, transactions, and integrations.
- Investigate account data, error codes, and transaction histories; reproduce issues in test accounts to distinguish software, configuration, and banking-related causes.
- Diagnose direct-deposit failures, bank-verification issues, payment holds, chargebacks, and bank-return/NSF codes.
- Guide customers through accounting and payroll workflows, including ROEs, vacation accruals, deductions, bank reconciliation, and project costing.
- Support QuickBooks Online data-migration workflows, including import troubleshooting and data validation.
- Document reproducible issues and coordinate Tier 2 or specialized-team escalations for suspected software defects and account investigations.
- Explain troubleshooting steps and software workflows clearly in English and French.

Mechatronics Workshop Instructor

Jan 2024 – Dec 2025

Skills Ontario Women's Initiative | Peterborough, ON

- Led hands-on 555-timer circuit workshops for Grade 8–10 students in groups of 20 or more, guiding assembly and troubleshooting.
- Explained electronics concepts and helped students connect circuit theory to practical results.

Research Assistant / Associate

Jul 2023 – Aug 2025

CAMIIT, Fleming College | Peterborough, ON | Part-time

- Supported testing and field deployment of STM32/ESP32-based smart-waste sensing devices using ultrasonic and NFC technologies.
- Performed SMD soldering and desoldering, hot-air rework, board cleaning, and conformal coating for electronics prototypes.
- Produced and finished 3D-printed components in SLS nylon, PLA, and resin; supported assembly, packaging, and delivery of parts.
- Trained technicians on equipment setup and troubleshooting during field deployments.

Engineering Intern

Aug 2022 – Sep 2022

Inyange Industries | Kigali, Rwanda

- Supported maintenance and automation tasks on manufacturing equipment, including boilers and Tetra Pak packaging lines.

ENGINEERING PROJECTS

Power of Induction – Mechatronics Capstone

Fleming College

- Assembled and wired the induction-heating circuit, soldered connections, and fabricated components for a team-developed flow-through water-heating prototype.
- Tested prototype heating performance, recording a water-temperature rise from approximately 21.9°C to 43°C in 35–40 seconds; troubleshooted tubing leaks and board overheating, and documented design limitations.

Swing-Wheel Parcel Sorter – Undergraduate Thesis

2021

- Created SolidWorks assembly and component drawings for a modular parcel-sorting mechanism with motor-driven wheels and directional adjustment.
- Documented transmission ratio, motor, belt drive, and bearing selection calculations; proposed a PLC and sensor-based control architecture.
- Delivered a design study with assembly views, sections, bills of materials, and detail drawings.

Visually Impaired Assistant – Personal Project

In Progress

- Rebuilding a college project that reached a working model, focused on indoor object detection and audio guidance.
- Prepared an indoor-object dataset and trained a YOLOv8 model; integration and evaluation of the rebuilt assistant remain in progress.

Compact 6:1 Two-Stage Spur Gearbox – Personal Project

In Progress

- Designing a two-stage spur gearbox (2:1 and 3:1 stages) with design calculations and Fusion 360 CAD.

EDUCATION

Postgraduate Certificate, Artificial Intelligence

2025

Fleming College | Peterborough, ON

Postgraduate Certificate, Mechatronics Engineering

2024

Fleming College | Peterborough, ON

B.Sc., Mechatronics Engineering

2021

Huzhou University | Zhejiang, China

AWARDS AND RECOGNITION

- Zhejiang Government Scholarship, two-time recipient
- Building Bridges for Women in Trades and Technology Scholarship